

Regime Conditioning Recovers Detection, Not Localisation: A Benchmark for Fault Onset on Load-Varying Fleet Telemetry

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CAOS open-research programme · CAOS_TruckVitals, regime-conditioned fault detection on fleet telemetry

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Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_TruckVitals

Abstract. A change detector on raw haul-truck telemetry spends most of its time detecting the haul cycle: strut pressure jumps when the tray is loaded, brake temperature climbs on every descent, fuel rate tracks the grade. The standard remedy, conditioning on the operating regime and monitoring within-regime residuals, is usually asserted rather than measured. It is measured here, with the confounds removed and the results that argue against it reported beside those that support it. A detector-free effect size on NASA C-MAPSS shows the mechanism directly: on the six-operating-condition subset FD002 the median fault signature is 0.16 pooled standard deviations but 11.95 within-regime standard deviations (ratios 90.2 on FD002 and 62.3 on FD004), and the single-condition subsets return exactly 1.00, a built-in negative control. At a matched event-counted false-alarm budget, detection falls from 0.93 to 0.17 (and 0.73 to 0.24) when operating conditions go from one to six, and regime-conditional residuals recover 0.95 (and 0.90), above the single-condition references; discovering the regimes by clustering recovers as much as being handed the labels. Three results temper this. Onset *localisation* is a null: chance-corrected skill differs by -0.08 ± 0.74 , paired over five seeds. No false-alarm reduction is found. And on a fourteen-rung detector ladder over a physically grounded synthetic fleet, conditioning helps six rungs, leaves six unchanged and hurts two learned novelty detectors, with a regime-label coverage of 0.998 ruling out missing regime labels as the cause. A prediction stated before two deep detectors were trained, that the shape of a detector’s statistic decides whether conditioning hurts it, is refuted: the boundary-shaped Deep SVDD is helped. Regime conditioning helps most detectors and hurts some healthy-only novelty models, and no rule established here predicts which.

Index Terms. Condition monitoring, change-point detection, operating regimes, fleet telemetry, haul trucks, statistical process control, novelty detection, pre-registration.

I. INTRODUCTION

A HAUL truck loaded on an uphill ramp and the same truck empty on the way down are not the same machine as far as a detector is concerned. Every monitored channel moves with payload, grade and speed, and none of that movement is a

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Interactive study and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_TruckVitals; the study is available at <https://truckvitals.fasl-work.com>. The detection engine is the separate PyPI package `regimecpd` (https://github.com/fsantibanezleal/CAOS_RegimeCPD).

fault. Raise the alarm threshold until the haul cycle stops firing and the fault you wanted sits under the threshold too. The remedy examined here is the standard one from multivariate statistical process control: learn the operating regimes from *context* channels (payload, grade, speed), and monitor each telemetry channel as a within-regime residual,

$$r_t = \frac{x_t - \hat{\mu}_{k(t)}}{\hat{\sigma}_{k(t)}}, \quad (1)$$

with $k(t)$ the regime at time t , the statistics estimated only on the unit’s own healthy baseline, and samples outside every seen regime left *unassigned* rather than snapped to the nearest one, because snapping lets a detector call a new operating condition a fault.

The contribution is a controlled measurement of this remedy rather than a new method: the confounds are removed by design, the null results are reported, and a counter-example is examined with a test whose prediction was stated before it was run. The detection engine is the separately published, MIT-licensed package `regimecpd`, and the conditions its implementation enforces are listed in Appendix A.

II. DATA, PROTOCOL, METRICS

A. Four data lanes, with declared limits

No public, redistributable dataset of continuous haul-truck telemetry with labelled faults was found; that absence shapes the design and is the main limitation. Four lanes stand in, each supporting a different part of the claim. (i) NASA C-MAPSS turbofan data [1] supplies the controlled experiment: subsets FD001/FD003 run one operating condition, FD002/FD004 run six, with the condition declared per sample, so regime variation can be switched on and off while everything else is held fixed. C-MAPSS records no onset instant; the onset of a unit is taken as the cycle at which its remaining useful life falls to 125 cycles, the usual cap in the prognostics literature, which is a convention rather than a measurement. (ii) The SCANIA APS dataset [2] supplies a decision problem under a published cost matrix (false positive 10, false negative 500). (iii) SCANIA Component X [3] supplies 23,550 real vehicles with a graded 5×5 cost matrix. (iv) A physically grounded synthetic haul

fleet (resistance-driven cycle, strut pressure from payload, TKPH-style tyre heating; the regime confound *emerges* from the cycle rather than being injected) supplies the only lane where the true onset instant exists by construction. Nothing in this lane is calibrated against a real truck.

B. The five protocol rules

(1) The raw arm and the residual arm reach every detector as the same kind of object, so the detector cannot know which it is scoring. (2) Arms are compared at a matched *false-alarm budget*, never at a matched threshold: a z-scored residual and a raw channel in bar live on different scales. (3) Alarms are *events* (rising edges of the thresholded statistic), not samples: a statistic that sits above threshold for an hour is one alarm an operator answers, not sixty. The event-counted rate is not monotone in the threshold, so the threshold search descends from the never-fires end; an ascending search can select a permanently alarming detector whose single rising edge looks like a perfect rate. (4) Uncertainty is bootstrapped over *units*, never over samples, which would manufacture significance by raising the sample rate. (5) Nothing is fitted on the data it is scored on: regime model and residual statistics fit on a healthy baseline window, the threshold is chosen on a disjoint calibration window as a *fleet* quantity, and detection is scored on the remainder.

C. Metrics

Detection rate counts faulty units whose first rising edge is at or after the onset; an earlier alarm is a false alarm, not an early detection. Delay is the median over detected units only, and is always printed beside the rate. Onset localisation error is the distance to the *nearest* retrospective changepoint, which is optimistic by construction, so it is always divided by the chance level for the same changepoint count, $S/(2(k+1))$ for k cuts over a span S : skill 1.0 means no better than scattering the same number of cuts at random.

III. THE MECHANISM, WITH NO DETECTOR IN IT

Before any detector enters, the mechanism itself can be measured as an effect size that no threshold rule, alarm convention or budget can move. For each C-MAPSS subset, take the median over channels of the fault signature measured in pooled standard deviations, and the same signature measured in within-regime standard deviations, on the healthy fleet baseline; the per-unit fault shift cancels in the ratio, so the ratio is a property of the operating-regime structure alone.

Fig. 1 shows the result. On FD002 the same physical fault is worth 0.16 pooled sigma, invisible, and 11.95 within-regime sigma, unmissable; on FD004, 0.17 and 10.17. The ratios are 90.2 and 62.3. The single-condition subsets return a ratio of exactly 1.00: with one regime, pooled and within-regime spreads are the same number. This built-in negative control confirms that the ratio measures the regime structure and nothing else.

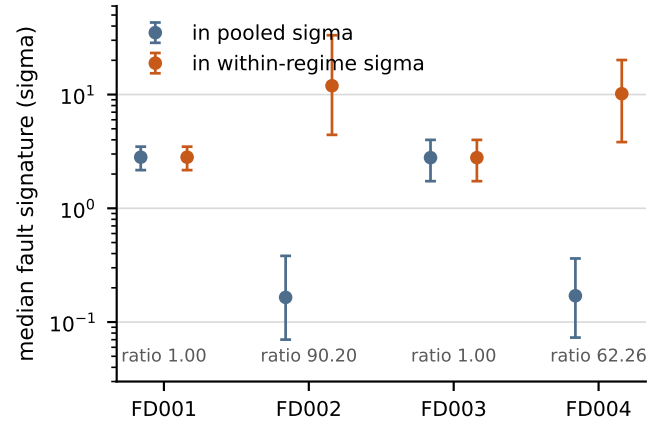


Fig. 1. The detector-free effect size. Median (p10–p90 over channels) fault signature per C-MAPSS subset, measured in pooled sigma (blue) and within-regime sigma (orange). Single-condition subsets (FD001, FD003) return a ratio of exactly 1.00, the built-in negative control; six-condition subsets return 90.2 and 62.3.

TABLE I
C-MAPSS DETECTION AT A MATCHED FALSE-ALARM BUDGET (CUSUM, ONE FALSE ALARM PER 1000 CYCLES). ARMS CONDITIONED ON A REGIME ARE BOLD. UNITS: FAULTY UNITS SCORED. GLOBAL-REGR IS A SINGLE GLOBAL CONTEXT REGRESSION WITHOUT SEGMENTATION: IT CONDITIONS ON CONTEXT BUT IS NOT A REGIME MODEL, SO IT IS REPORTED SEPARATELY.

Pair	Arm	Detection	Regimes	Units
FD001/FD002	raw, 1 condition (ref.)	0.93	1	28
	raw, 6 conditions	0.17	–	87
	residual, observed	0.95	6	87
	residual, clustered	0.95	6	87
	global-regr	0.89	1	87
FD003/FD004	raw, 1 condition (ref.)	0.73	1	52
	raw, 6 conditions	0.24	–	146
	residual, observed	0.90	6	146
	residual, clustered	0.90	6	146
	global-regr	0.88	1	146

IV. DETECTION UNDER REGIME VARIATION

Table I gives the controlled contrast: the same CUSUM detector ($k=0.5$), the same healthy data per arm, thresholds cross-fitted over units, a budget of one false alarm per 1000 cycles, and two confounds removed by design. FD002 carries more informative channels than FD001, and the statistic is a maximum across channels, so more channels raise the false-alarm rate on their own; only the channels informative in both subsets of a pair are used (15 and 16 on the two pairs). Fraction-based fit windows drop units at different rates in different subsets, so the fit and calibration windows are absolute cycle counts (60 and 30 cycles); units whose onset falls inside them are not scored.

Three readings. First, regime variation alone destroys detection (from 0.93 to 0.17; from 0.73 to 0.24), and conditioning recovers it to *above* the single-condition references (0.95; 0.90), at a lower realised false-alarm rate than the raw arm. Second, *discovering* the regimes by k -means on the context channels recovers exactly as much as being handed the declared condition labels (0.954 equals 0.954 to three decimals):

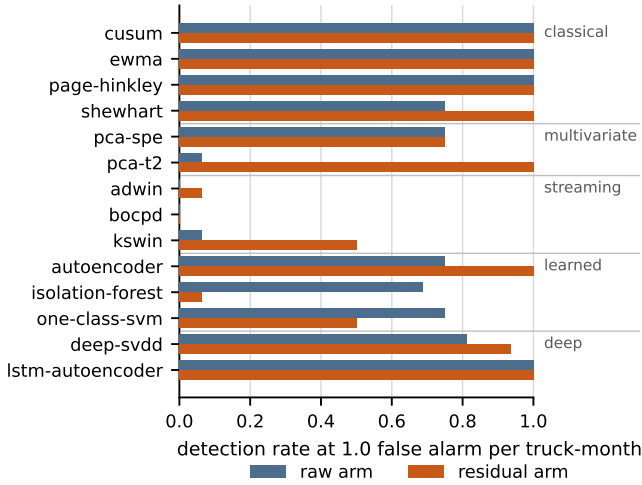


Fig. 2. The fourteen-rung ladder at a matched budget, by tier. Conditioning helps six rungs (largest: Hotelling T^2 , from 0.06 to 1.00), leaves the detection rate of six unchanged (four at ceiling, SPE at 0.75, BOCPD at zero) and hurts two learned novelty models. BOCPD detects nothing on either arm (gradual ramps against an abrupt-change model); ADWIN’s integer statistic leaves it only $d+1$ operating points.

the labels carry no information the context channels do not. Third, a global context regression with no segmentation at all recovers most of the gap (0.89; 0.88); on FD004 the margin between conditioning and plain regression is 0.02, too small to count as an advantage.

V. THE LADDER, AND A COUNTER-EXAMPLE

On the synthetic fleet (36 units, 16 faulty, four fault families injected with known onset), fourteen online detectors run on both arms at the same budget of 1.0 false alarm per truck-month. Twelve form the ladder proper: Shewhart [4], CUSUM [5], EWMA [6], Page-Hinkley [5], [7] (reported as a CUSUM variant, not an independent baseline; at equal allowance the two correlate above 0.98), Hotelling T^2 and PCA SPE [8], [9], [10], BOCPD [12], KSWIN [14], ADWIN [13], isolation forest [17], one-class SVM [18] and a dense autoencoder [19]. Two deep detectors, Deep SVDD [22] and an LSTM encoder-decoder [23], test an explanation of the counter-example below (Section VI). PELT [15] and the multidimensional matrix profile [16] are *excluded*: both are retrospective (the profile value at a window is computed against matches anywhere in the record, including after it), and an online delay metric must never score a method that has read the future. A deliberately generous trivial baseline, a fixed threshold on the single channel each fault moves first, *told the answer*, scores 0.75 at 123 minutes; a ladder that cannot beat it would mean the lane is too easy to be informative.

Fig. 2 and Fig. 3 carry the result. The largest effect on the ladder is the mechanism made visible: raw T^2 detects 1 of 16 faulty trucks, because the retained principal subspace of a haul truck is dominated by the duty cycle, and a fault that moves the machine within that structure looks like a load change; the same statistic on residuals detects 16 of 16 at a $3.6\times$ shorter delay. SPE sits at 0.75 on both arms, and its gain is delay

TABLE II
THE PREDICTION RECORDED BEFORE TRAINING, AGAINST THE MEASURED DETECTION RATES.

Rung	Predicted	Raw	Residual
deep-svdd (boundary)	hurt	0.812	0.938
lstm-autoencoder (reconstruction)	not hurt	1.000	1.000

(from 210.5 to 86 minutes): the quarter of faults it misses never breaks the correlation structure at this severity.

The main negative finding is a counter-example. Conditioning *hurts* the isolation forest (from 0.69 to 0.06) and the one-class SVM (from 0.75 to 0.50 on detection rate; its shorter conditioned delay is measured on the half it still catches). Unassigned regime samples destroying the ten-sample feature windows these models score would explain it, but regime coverage on the residual arm is 0.998, so there are almost no gaps to blame. Other learned rungs are helped: the dense autoencoder (from 0.75 to 1.00, from 174 to 90 minutes) and Deep SVDD (from 0.81 to 0.94), while the LSTM encoder-decoder is at ceiling on both arms.

VI. TESTING AN EXPLANATION OF THE COUNTER-EXAMPLE

A candidate explanation is that the *shape* of a novelty detector’s statistic decides whether conditioning helps it. A boundary-shaped detector learns where the healthy cloud is: on the raw arm that cloud is a few tight regime clusters, and anything off-cluster is easy to isolate, while conditioning replaces them with one roughly isotropic cloud inside which a moderate shift stays. A reconstruction-shaped detector scores error through a rank-limited map, so a fault that breaks cross-channel structure produces error however isotropic the healthy cloud looks. The explanation predicts that conditioning hurts boundary-shaped detectors and does not hurt reconstruction-shaped ones.

The prediction was recorded before two deep detectors were trained, one of each shape and neither from the implementation families of the rungs that suggested it: Deep SVDD [22], boundary-shaped (it learns a minimum-volume hypersphere around the healthy representations), and an LSTM encoder-decoder [23], reconstruction-shaped. Table II gives the outcome. Deep SVDD is *helped*, by 0.125, well beyond the 0.01 margin used here to call a direction, which refutes the prediction. The LSTM result is nearly uninformative: it sits at ceiling on both arms and could not have been hurt from there, although its median delay falls from 170 to 54 minutes.

The measurements stand: the isolation forest (from 0.688 to 0.062) and the one-class SVM (from 0.750 to 0.500) are hurt. The explanation does not: three boundary-shaped rungs split two hurt against one helped, which is not a rule. One visible difference, that Deep SVDD learns its representation while the two hurt rungs partition a fixed input space, would need its own prediction, recorded before its own test, and remains an open question.

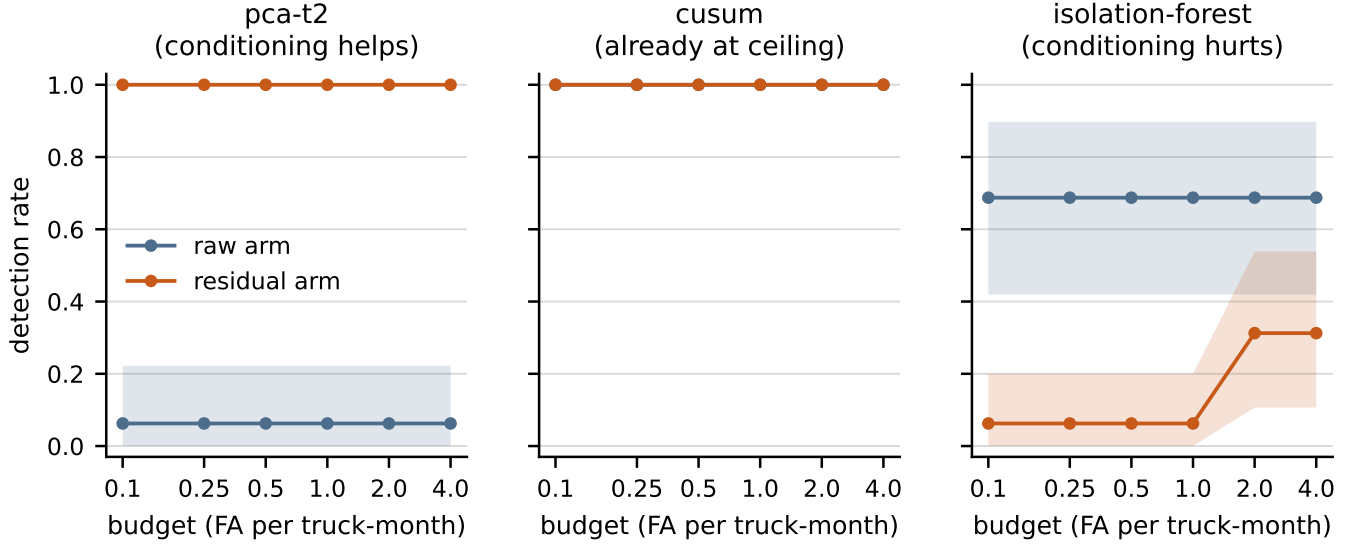


Fig. 3. Alarm-budget curves with bootstrap-over-units intervals for a rung conditioning helps (Hotelling T^2), a rung already at ceiling (CUSUM), and a rung conditioning hurts (isolation forest). The single-point ladder of Fig. 2 is one slice of these curves; a method that wins only at one budget has not won.

TABLE III
ONSET LOCALISATION VIA NEAREST RETROSPECTIVE CHANGEPOINT (36 UNITS, SEED 0; MEDIAN). SKILL IS CHANCE/ERROR; 1.0 MEANS NO BETTER THAN RANDOM CUT PLACEMENT.

Arm	Error (min)	Cuts	Chance (min)	Skill
raw	121.5	2	236.2	1.94×
residual	52.5	13	50.6	0.96×

VII. LOCALISATION IS A NULL

Retrospective PELT segmentation estimates the onset instant on both arms; Table III shows why the chance column is the number to read. The conditioned arm has the smaller absolute error on every one of five seeds, by factors of 2.3 to 6.8, and all of that advantage is explained by producing 11.5 to 14 changepoints against 2 to 3: at thirteen cuts the chance level is already 50.6 minutes. Chance-corrected, the paired skill difference is -0.08 ± 0.74 , with conditioning ahead in two of five seeds, and the per-seed skill ranges from 0.87 to 1.94 on the raw arm and from 0.86 to 1.75 on the residual arm. Regime conditioning does not reliably improve onset *localisation*; this does not touch the detection result, which is a different measurement on different data, stable across detectors and subset pairs.

No false-alarm reduction is found. C-MAPSS measures detection at a *fixed* false-alarm budget. On the synthetic lane neither arm raises a false alarm on the scored data at budgets up to 1.0 per truck-month, and at 4.0 the residual arm realises more false alarms than the raw arm on six rungs and fewer on two.

VIII. TWO FINDINGS NOT ABOUT REGIMES

The cost lanes produced two results even though no regime enters either. On SCANIA APS, with the dataset’s own cost

matrix (FP 10, FN 500) and thresholds chosen out-of-fold, the F1-optimal decision rule attains the best F1 (0.807) and a total cost of 47,890, while the cost-optimal rule attains 11,670 at a worse F1 (0.652): optimising the metric almost everyone reports selects a decision $4.1\times$ more expensive than optimising the one the operator pays. On Component X, minimising expected cost under the graded matrix helps an unweighted model and degenerates to flagging every vehicle on a class-weighted one: class weights and a cost matrix encode the same asymmetry through two mechanisms, and applying both double-counts it, because an expected-cost decision needs calibrated probabilities and a class-weighted model’s are not. Never flagging anything scores 97.3% accuracy and the worst cost of all, which is the same lesson from the other side.

IX. RELATED WORK

The components are deliberately standard: control charts and their multivariate extensions [4], [5], [6], [8], [9], [10], [11], Bayesian and windowed online change detection [12], [13], [14], retrospective segmentation [15], matrix profiles [16], healthy-only novelty models [17], [18], [19], and conformal calibration as the engine-level route to budget-comparable scores [20], [21]. Conformal calibration is implemented in the engine, but the numbers here use a direct event-counted threshold search, so no conformal guarantee applies to them. Closest in intent, Carpentier et al. [24] model heavy-duty trucks “contextually”, but their context is a per-vehicle *cohort*: the fleet is clustered by specification and one model is trained per cohort, assigned once per vehicle. There is no per-timestep regime label and no residual channel. Their approach partitions the fleet; this one partitions the timeline, and the two compose without interacting.

X. LIMITATIONS

The truck-specific results are synthetic: magnitudes are physically plausible for a large rigid hauler and are not measurements, and the missing real lane is the study's largest limitation. The controlled detection experiment lives on turbofans, chosen because the regime structure is declared and switchable, not because turbofans resemble trucks, and its onset is a convention. The ladder runs at one severity and one seed (the seed sensitivity of the localisation result is why it is a null); the budget sweep covers 0.1–4.0 alarms per truck-month. ADWIN's integer statistic makes it structurally ill-suited to a common budget, and its rows understate what a confidence sweep might show. The learned-tier counterexample rests on one lane and one feature contract, and no explanation of it is established.

XI. CONCLUSION

On load-varying telemetry, regime variation alone can destroy budgeted detection, and conditioning on regimes learned from context channels recovers it to above single-condition references, with clustering as good as declared labels. The same conditioning does not reliably improve onset localisation, yields no false-alarm reduction, and hurts two learned novelty detectors while helping position and accumulation detectors. Which detectors it hurts is predicted by no rule established here: the statistic-shape explanation was refuted by a test whose prediction was recorded before it was run (Section VI). In practice, control charts and T^2 should be run on the residual arm, novelty detectors kept on the raw arm or re-validated after conditioning, and a localisation figure trusted only beside its chance level.

APPENDIX A IMPLEMENTATION SAFEGUARDS

The numbers above are sensitive to implementation details that, when wrong, produce plausible numbers rather than errors. The engine (`regimecpd`) and the product pipeline enforce the following conditions, each covered by a regression test. A channel constant to within numerical noise is not standardised by that noise, in the regime residuals and in the linear residual branch alike, since a noise-scale denominator lets a single unit set the fleet threshold. The threshold search descends from the never-fires end and considers every candidate threshold, because the event-counted false-alarm rate is not monotone in the threshold. A missing-value gap inside a sustained excursion does not start a new alarm, which would otherwise bias the residual arm. The scale floor of the novelty detectors uses an unsigned scale, since residual means are zero by construction. ADWIN's cut threshold follows the bound of [13]. Both arms receive the same healthy data, and thresholds are chosen on a calibration window disjoint from the scored data.

APPENDIX B REPRODUCTION

Every number is reproduced from the repository artifacts under `data/artifacts/:` `cmapss_mechanism.json`

and `cmapss_regime_contrast.json` (the C-MAPSS mechanism and contrast), `synthetic_benchmark.json` (the ladder, the budget curves and the trivial baseline), `onset_seed_sweep.json` (localisation over five seeds), `aps_cost.json` and `componentx.json` (the cost lanes). They are produced by the pipeline under `data-pipeline/` with the `regimecpd` engine, and the figure script under `manuscripts/regime-conditioning-benchmark/figures/` reads them.

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